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MILITARY SPECIFICATION

PROJECTILE, HEAT-T, 76MM, M495
METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Projectile, HEAT-T, 76MM, M495 Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-M-6867	- Magnetic Inspection Units
MIL-I-6868	- Inspection Process, Magnetic Particle
MIL-C-60536	- Coating, Anodic, Hard for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-410	- Certification of Magnetic Particle Inspection Personnel
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition

FSC 1315

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2.1 (Cont'd)

MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Armament Command

F8597471	- Metal Parts Assembly
LI-8597471	- Index Listing for Inspection Equipment for Projectile, HEAT-T, 76MM, M495, Metal Parts Assembly
F7548604	- Pallet for Fixed and Semifixed Shell Assembly, Details, Packing and Marking

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1	- Surface Texture, Surface Roughness, Waviness and Lay

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

ASTM Method E-8	- Tension Testing of Metallic Materials
ASTM Method E-10	- Brinnell Hardness of Metallic Materials
ASTM Method B117	- Salt Spray (FOG) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

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3.2 Heat treatment or stress relief.-The heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in any component the mechanical properties specified on the applicable component drawing.

3.3 Wire assembly.

3.3.1 Pull test.-The wire and eyelet assembly shall withstand a pull test of 125 pounds, minimum.

3.3.2 Fuze wire terminal.-The fuze wire terminal when crimped to the fuze wire shall withstand a pull test of 20 pounds, minimum.

3.3.3 Insulation breakdown.-The wire assembly when assembled to the metal parts assembly, shall exhibit no evidence of electrical breakdown when subjected to a direct current potential differential of 1500 volts.

3.3.4 Electrical continuity.-The wire assembly when assembled to the metal parts assembly shall exhibit no evidence of electrical discontinuity.

3.3.5 Electrical resistance.-After assembly, the resistance between the forward end of the spike and the rear of the body shall not exceed 0.2 ohm, max.

3.4 Protective coating.-Protective coating shall be in accordance with the applicable drawings and specifications. When protective coating is required by component details, finish requirements will apply to all indicated surfaces of the component regardless of contour configuration.

3.5 Painting.-Painting shall comply with the requirements on the applicable drawings. All paint and marking shall be dry to the touch before testing and packing for shipment.

3.6 Air pressure.-The explosive charge cavity of the assembly shall be capable of containing an internal air pressure of 3 $\pm$ .5 psi without evidence of leakage, for a minimum of 30 seconds.

3.7 Metal defects.

3.7.1 Components.-All components, and component cavities, shall be free from cracks, splits, cold shuts, inclusions, pipe, porosity, bursts, folds, seams and other metal defects.

3.7.2 Component cavities.-In addition to 3.7.1, all component cavities shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

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3.7.3 Visual standards.-Visual standards for metal defects will be established for each contract after the first three month's production. However, the establishment of visual standards will not waive the intent of the requirement for compliance with 3.7.1 and 3.7.2. (See 6.2)

3.8 Proving ground functional tests.-The projectile metal parts shall reveal no evidence of metal parts failure when tested at excess and service pressure. Metal parts failure shall be:

a. Loss or breakup of metal parts components in the gun bore or in flight.

b. Penetration, Contained in Appendix (See 6.6).

3.9 Workmanship.-The requirements for workmanship are as specified on the applicable drawing, referenced specifications and the following:

3.9.1 Material defects.-Molded components shall be free of porosity and excessive flash.

3.9.2 Hydrogen embrittlement.-Copper cones which are annealed during manufacture shall show no evidence of gassing or open grain structure characteristic of hydrogen embrittlement.

3.9.3 Surface finish.-The requirements for surface finish are as detailed on the applicable drawing.

3.9.4 Threads.-Threads shall be full and undamaged for the entire minimum length as required by the applicable drawings.

3.9.5 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.9.6 Foreign matter.-No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter prior to inspection and shipping.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.

a. Successful passage of all inspection required by the drawings and specifications shall be considered as a condition precedent to performing the Proving Ground functional tests.

b. Acceptance of a lot by the Government shall be contingent on successful passage of the Proving Ground Functional tests.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 50 metal parts assemblies as follows:

35 metal parts assemblies, complete, including paint
5 metal parts assemblies, complete except paint
5 sets of sub-assemblies
5 sets of component parts unpainted, unassembled

Thirty of the above complete and painted metal parts assemblies shall be forwarded to a designated Government facility for loading and subsequent ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply. A first article sample or portion thereof, as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing processes, materials used, drawings, specifications or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample will be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

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4.2.1.3 Abrasion resistance.

4.2.1.4 Test panels.-When the protective coating is being applied to the initial production sample of the Fin Assembly (or components thereof), 4 test panels as specified in paragraph 9.4.3 of MIL-C-60536(MU) shall be processed along with the product, through each step in the coating process. After coating the 4 panels will be forwarded to a Government approved laboratory for testing in accordance with 9.4.3 of MIL-C-60536(MU) to determine compliance with paragraph 8.4.1 of MIL-C-60536(MU).

4.2.1.5 File test.-At the same time the test panels provided by 4.2.1.4 are being subjected to the abrasion test at the Government approved laboratory 2 fin assemblies (or two sets of component parts thereof) from the initial production sample shall be subjected to the file test in accordance with paragraph 9.4.4 of MIL-C-60536(MU) to establish correlation (of hardness of coat) between the panels from 4.2.1.4 and the file test of the fin assemblies or components thereof. The file test will be performed using the equipment and procedures that will be used in production testing.

4.2.1.6 Correlation.-When performing the file test prescribed by 4.2.1.5, the following information shall be obtained:

a. The type and model of the filing machine used.
b. The type of file (roughness) used.
c. The length of the file stroke.
d. The pressure applied to the file.
e. The number of strokes of the file required to expose the basic metal.

The information thus obtained shall be the basis for determining compliance of the production sample herein after prescribed, wth the requirements of paragraph 8.4.2 MIL-C-60536(MU).

4.3 Inspection provisions, test methods and procedures. (See 6.3)

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Mechanical properties.

4.3.1.2.1 Hardness and tension test.

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4.3.1.2.1.1 Heat treatment lots (definition).-Heat treatment lots shall contain spikes and bodies from not more than one heat of steel, that has been heat treated in the same heat-treating equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.2.1.2 Heat treatment lots (Hardness).-In order to determine uniformity of heat treatment, each body and spike in the heat treatment lot (as defined above) shall be subjected to a hardness test with a Brinnel Hardness Tester, using a 5 second minimum dwell or by any other method of hardness testing approved by the procuring activity.

4.3.1.2.1.3 Heat treatment lots (Tension test sample).-Two bodies and two spikes (as applicable) shall be selected from each heat treatment lot, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of one or more tension test specimens to comply shall be cause for rejection of the lot of components represented by the failing specimen. On tangentially transverse specimens the elongation shall be recorded for information only. At the option of the supplier a production control may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Bodies or spikes which fall outside of the established range may be re-heat treated to fall within the range, or mechanical properties may be determined from newly selected samples which will represent the true extremes of hardness in the heat treatment lot.

4.3.1.2.1.4 Stress relieved lots (definition).-Stress relieved lots shall contain spikes and bodies cold shaped by the same process, utilizing the same cold shaping equipment (as applicable) and stress relieved in one unchanged process. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.2.1.5 Stress relieved lots (Hardness).-Twenty-five cold shaped bodies and spikes (as applicable) from the first batch of each stress relieved lot (as defined above) processed through the stress relief furnace and five bodies or spikes from each 1 hour production thereafter for the balance of the stress relieved lot shall be subjected to a hardness test to assess uniformity of the product.

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4.3.1.2.1.6 Stress relieved lots (Tension test samples).-Two bodies and two spikes (as applicable) shall be selected from each stress relieved lot, one of the hardest and one of the softest, as determined by the hardness test from the first twenty-five bodies and spikes shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot. On tangentially transverse specimens the elongation shall be recorded for information only. When it has been determined that the mechanical properties comply and the furnace temperature and time has been established, the furnace temperature and time shall be maintained throughout each stress relief lot. The furnace temperature may vary within twenty-five degrees fahrenheit but not below the minimum temperature of 700 degrees fahrenheit and the minimum 30 minutes soak time. Furance temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated each 8 hours by an independent method.

4.3.1.3.1.7 Miscellaneous lots (definition).-Small lots (such as spikes or bodies manufactured from abnormally small heats of steel or spikes or bodies that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment or stress relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with 4.3.1.2.1.1 or 4.3.1.2.1.4 as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.2.1.8 Miscellaneous lots (Hardness).-Miscellaneous heat treatment lots of spikes or bodies shall be hardness tested in accordance with 4.3.1.2.1.2. Miscellaneous cold shaping lots of bodies shall be hardness tested in accordance with 4.3.1.2.1.5.

4.3.1.2.1.9 Miscellaneous lots (Tension test sample).-Samples for tension test of miscellaneous heat treatment or stress relieved lot of bodies or spikes shall be selected and tested in accordance with 4.3.1.2.1.3 or 4.3.1.2.1.6 as applicable. Failure of one or more tension test specimens to comply with mechanical properties shall be cause for rejection of the lot of components represented by the failing specimen. On tangentially transverse specimens, the elongation shall be information only. Tension tests shall be performed on each miscellaneous heat of steel before being grouped into a miscellaneous lot.

4.3.1.2.1.10 Hardness test method.-After final heat treatment or stress relief, but prior to finish machining, one Brinnel hardness test shall be performed on the large outside diameter of the spike and on the rear face of the body. Prior to performing the test, the area in which the readings are to be taken shall be cleaned of scale or other foreign matter. The test shall be performed using equipment and procedures prescribed by ASTM Method E-10.

4.3.1.2.1.11 Tension test methods.

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4.3.1.2.1.11.1 Body.-From the samples provided in accordance with the heat treat, stress relieved or miscellaneous lots, two tension test specimens conforming to the largest obtainable standard "R" type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of the body wall. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements and with the approval of the procuring agency, the Government shall not be restricted from testing additional longitudinal specimens at other locations to determine that the mechanical properties of the body meet the drawing requirements. The criteria for performing this test shall be prescribed by ASTM Method E-8, except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.3.1.2.1.11.2 Spike.-From the samples provided in accordance with the heat treat, stress relieved or miscellaneous lots, two tension test specimens conforming to the largest obtainable standard "t" type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of the stem portion of the spike. In addition, two tension test specimens conforming to the largest obtainable standard "R" type as prescribed by ASTM Method E-8 shall be taken in a tangentially transverse direction from opposite sides at the midpoint of the flange area. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements and with the approval of the procuring agency, the Government shall not be restricted from testing additional specimens at other locations to determine that the mechanical properties of the spike meet the drawing requirements. The criteria for performing this test shall be as prescribed by ASTM Method E-8, except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.3.1.2.1.12 Fins (Tension test samples).-Two fins from each lot of 1200 fins or from 400 feet of extrusion or fraction thereof shall be subjected to the tension test. Failure of any specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot represented by the sample.

4.3.1.2.1.13 Fin (Tension test method).-From the samples proveded in accordance with 4.3.1.2.1.12, one tension test specimen conforming to the largest obtainable standard "R" type specimen shall be taken longitudinally from each component. The criteria for performing this test shall be as prescribed by ASTM Method E-8.

4.3.1.2.1.14 Boom (Tension test samples).-Two booms from each lot of 1200 or fraction thereof shall be subjected to the tension test. Failure of any specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot represented by the sample.

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4.3.1.2.1.15 Boom (Tension test method).-From the samples provided in accordance with 4.3.1.2.1.14, one tension test specimen conforming to the largest obtainable standard "R" type specimen shall be taken longitudinally from each component. The criteria for performing this test shall be as prescribed by ASTM Method E-8.

4.3.1.2.1.16 Hydrogen embrittlement sample.-After final annealing, one copper cone from each lot of 5000 cones shall be examined for hydrogen embrittlement. Failure to comply with the requirement of 3.9.2 shall be cause for rejection of the lot represented by the sample. (See 6.4)

4.3.1.2.1.17 Hydrogen embrittlement test method. From the sample supplied in accordance with 4.3.1.2.1.16, two specimens, one from the front (large) end and one from the rear small end, shall be removed for micro structure study. Specimens shall be polished, etched if desired, and examined for embrittlement under a microscope of not less than 75 diameters.

4.3.1.2.2 Magnetic particle inspection.-After heat treatment or stress relief and finish machining, each body shall be subjected to a magnetic particle inspection. Failure of any body to comply with 3.7 shall be cause for rejection of the body.

4.3.1.2.3 Magnetic particle (Inspection method).-The procedures for performing this test utilizing the wet fluorescent method shall be as specified in MIL-I-6868, for indirect magnetization by either the continuous method or the residual method, utilizing a circular magnetic field. The suspension shall be applied to the entire body, inside and outside. After draining, the body shall be throughly inspected in a darkened area under an ultra-violet light which will produce adequate fluorescence of the magnetic particles. The solid content of the suspension shall be maintained at a level recommended by the suspension manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by specification MIL-I-6868. When performing the test for solid content of the suspension, if it is determined that it does not comply, all projectile bodies that have been tested and accepted since the last satisfactory test was performed shall be re-tested after the suspension has been brought to the proper level. Magnetic particle inspection shall be performed on equipment that complies with the requirements of MIL-M-6867. The operator(s) of the equipment and inspector(s) making the determinations shall be certified in accordance with MIL-STD-410.

4.3.1.2.4 Torque test samples.-Five fin and boom assemblies and five projectile sub-assemblies from each one hour's production or fraction thereof from each torque machine in operation shall be selected at random and tested to determine compliance with the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the production represented by the sample.

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4.3.1.2.5 Fin and boom torque test method.-The fin and boom assemblies supplied in accordance with 4.3.1.2.4 shall be mounted in an approved holding device in such a manner that the boom is prevented from turning. The required torque shall be applied to the fin gradually in the direction required for assembly. Any perceptible movement of the fin shall be cause for rejection of the fin and boom assembly. Testing shall be conducted while thread sealant is wet.

4.3.1.2.6 Projectile sub-assembly torque test method.-The projectile sub-assemblies supplied in accordance with 4.3.1.2.4 mounted in an approved holding device in such a manner that the body is prevented from turning. The torque shall be applied to the spike in the direction required for assembly. Any perceptible movement of the spike shall be cause for rejection of the metal parts assemblies. Testing shall be conducted while thread sealant is wet.

4.3.1.2.7 Strength of wire assembly samples.-Ten wire and insulator assemblies from each lot of 5000 or fraction thereof shall be subjected to a pull test to determine compliance with 3.3.1. Failure of one or more assemblies to comply shall be cause for rejection of the lot represented by the sample.

4.3.1.2.8 Strength of wire assembly method.-The samples selected in accordance with 4.3.1.2.7 shall be placed in an approved holding fixture and the load applied gradually in the direction of the wire axis. Assemblies tested shall not be used in metal parts assemblies.

4.3.1.2.9 Air pressure test.-Each metal parts assembly consisting of body lock ring, conduit and cone, components as a minimum shall be subjected to an air pressure test to determine compliance with 3.6. Failure of any assembly to comply shall be cause for rejection of that assembly.

4.3.1.2.10 Air pressure method.-The assembly shall be mounted in an approved holding fixture equipped with a gage to visibly determine the pressure applied and held for the minimum time required. A drop in pressure reading prior to the expiration of the required time will indicate a leaking assembly.

4.3.1.2.11 Insulation breakdown test.-Each metal parts assembly shall be subjected to this test to determine compliance with 3.3.3. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.1.2.12 Insulation breakdown test method.-The direct current shall be applied between the fuze wire terminal and the rear of the body.

4.3.1.2.13 Electrical continuity test.-Each metal parts assembly shall be subjected to this test to determine compliance with 3.3.4. Failure of any assembly to comply shall be cause for rejection of the assembly.

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4.3.1.2.14 Electrical continuity test method.-The voltage shall be applied between the eyelet terminal and the fuze wire terminal and the circuit monitored for evidence of discontinuity.

4.3.1.2.15 Electrical resistance test.-Each metal parts assembly shall be subjected to this test to determine compliance with 3.3.5. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.1.2.16 Electrical resistance test method.-The resistance between the forward end of the spike and the rear of the body shall be checked with an ohmeter capable of performing this test accurately and consistently.

4.3.1.2.17 Fuze wire terminal test.-The fuze wire terminals of 25 metal parts assemblies from each 8 hours production or fraction thereof shall be subjected to a pull test to determine compliance with 3.3.2. Failure of any terminal to comply shall be cause for rejection of the lot of metal parts assemblies represented by the sample.

4.3.1.2.18 Fuze wire terminal test method.-The samples selected in accordance with 4.3.1.2.17 shall be tested using an approved holding fixture and the load applied gradually to the direction of the wire axis.

4.3.1.2.19 Protective coating.

4.3.1.2.19.1 Metal parts assembly sample.-The sample size and acceptance criteria for the inspection of cleaning the metal parts assembly and the application of the protective coating shall be in accordance with the requirements of the applicable drawing and the specifications referenced thereon. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

4.3.1.2.19.2 Metal parts assembly test method.-The procedures for testing the cleaning of assemblies and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specification listed thereon. The salt spray test shall be performed utilizing equipment prescribed by ASTM Method B-117.

4.3.1.2.19.3 Abrasion resistance test sample.-One fin assembly (or set of components thereof) from each 8 hours production or fraction thereof from the protective coating process shall be subjected to this test to determine compliance with paragraph 9.4.4 of MIL-C-60536(MU), except that the information obtained during file test in 4.2.1.6 shall be utilized as indicated in 4.2.1.5. Failure of the assembly or components thereof to comply shall be cause for rejection of the product processed since the last acceptable test was performed.

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4.3.1.2.19.4 Abrasion resistance file test method.-The samples provided by 4.3.1.2.1.19.3 shall be tested using a filing machine of a type and model that was used in establishing correlation as indicated in 4.2.1.5. The samples shall be tested using the same type (roughness) file, the same length of stroke, the same number of strokes and the same file pressure as was used on the initial production sample.

4.3.1.3 Proving Ground functional tests.-The following Proving Ground tests will be performed by the Government:

4.3.1.3.1 First article sample.

4.3.1.3.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated.

TABLE I

Phase	Quantity
Excess pressure	10
Service pressure	10
Service pressure (Penetration, Contained in Appendix (See 6.6).	10

4.3.1.3.2 Production lot sample.-The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

Phase	Quantity
Excess pressure	10 each lot
*Service pressure	10 each lot

* After five (5) consecutive lots have been ballistically tested and accepted, ballistic testing may be discontinued, with the approval of the procuring agency. However, implementation of ballistic testing will be instituted whenever directed by the procuring contracting officer.

4.3.1.3.3 Excess pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.8 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

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4.3.1.3.4 Service pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.8 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.1.3.5 Excess pressure method.-The metal parts assemblies supplied in accordance with Table I and II for these phases shall be inert loaded to the service weight and assembled into complete rounds and fired in a gun for which the round is standard. The propellant charge will be adjusted to obtain 109 to 117 percent of the upper pressure limit of the propellant used with the applicable projectile. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.8.

4.3.1.3.6 Service pressure method.-The metal parts assemblies supplied in accordance with Tables I and II for this phase shall be inert loaded to the service weight and assembled into complete rounds and fired in a gun for which the round is standard. The rounds will be fired to impact on a vertical target placed 2000 meters from the gun muzzle. The test rounds will be fired alternately with the projectiles from a referenced lot for comparison of velocity-accuracy at this range. Accuracy in terms of maximum spread in inches, both horizontal and vertical and probable error (PE) in mils along with velocity and pressure of each round fired will be recorded for information purposes only. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.8.

4.3.1.3.7 Penetration phase.-The metal parts assemblies supplied in accordance with Table I for this phase shall be loaded to the service weight and assembled into complete rounds and fired in a gun for which the round is standard. At a target as specified in the Classified Appendix. The target shall be placed 300 feet from the gun muzzle at an obliquity of 60 degrees. Penetration shall comply with the requirements specified in the Classified Appendix. (See 6.6)

4.3.1.4 Inspection equipment.-Index of Inspection Equipment List Number LI-8597471 identified the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation specified on the applicable LI.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity (See 6.5). An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

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4.3.2.1 Tool control dimensions.-Dimensions marked tool control shall be gaged during First Article testing and after each 250 items produced on each machine in operation and after any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.3.2.2 Spike, Drawing D8597479, covering a detail of Drawing F10535894.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Runout of bell housing with pitch diameter of large thread and flange	Gage
102. True position of inside diameter of stem with pitch diameter of small thread	Gage
103. Pitch diameter of large thread, min.	Gage
104. Major diameter of large thread, min.	Gage
105. Diameter of stem (inside)	Gage
106. Diameter of stem (outside)	Gage
107. Diameter of insulator cavity	Gage
108. Diameter of small thread undercut	Gage
109. Depth of stem cavity from rear face	Gage
110. Length to forward basic diameter (bell housing)	Gage
111. Length to rear basic diameter (bell housing)	Gage
112. Length of thread pilot on stem	Gage
113. Length from front face to stem shoulder	Gage
114. Length, total	Gage
115. Width of small thread undercut	Gage
116. Variation in wall thickness of stem	Gage
MINOR	
201. Diameter of large thread undercut	Gage
202. Length from rear face to front of flange	Gage
203. Length from rear face to rear of flange	Gage
204. Width of large thread undercut	Gage
205. Radii or chamfer missing or incorrect	Visual (See 6.1)

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
206. Surface finish improper	Visual (See 6.1)
207. Burr	Visual
208. Foreign matter	Visual

4.3.2.3 Cone, Drawing C8597478, covering a detail of Drawing F10535894.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Parallelism of flange	Gage
102. Diameter of flange	Gage
103. Diameter inside (rear)	Gage
104. Diameter outside (rear)	Gage
105. Diameter of conduit hole	Gage
106. Length to forward basic diameter	Gage
107. Length to rear basic diameter	Gage
108. Length, total	Gage
109. Location of conduit hole	Gage
110. Thickness of wall (slant length)	Gage
111. Thickness of wall (rear face)	Gage
112. Thickness of flange	Gage
113. Variation in wall thickness, circumferential	Gage
114. Variation in wall thickness slant length	Gage
MINOR	
201. Radii or chamfer missing or incorrect	Visual (See 6.1)
202. Surface finish improper	Visual (See 6.1)
203. Burr	Visual
204. Foreign matter	Visual

4.3.2.4 Body, Drawing D8597476, covering a detail of Drawing F10535894.

CRITICAL	
1. Metal defective	Visual (Probe light)
MAJOR	
101. Runout of charge cavity with bourrelet (front)	Gage
102. Runout of charge cavity with bourrelet (rear)	Gage

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4.3.2.4 (Cont'd)

	Defect	Method of Inspection
103.	True position of slots with fuze cavity	Gage
104.	True position of major diameter of rear thread with fuze cavity	Gage
105.	Pitch diameter of spike thread, max.	Gage
106.	Minor diameter of spike thread, max.	Gage
107.	Diameter of charge cavity (front)	Gage
108.	Diameter of charge cavity (rear)	Gage
109.	Diameter of outside basic length (rear)	Gage
110.	Diameter of outside basic length (front)	Gage
111.	Diameter of bourrelet, min.	Gage
112.	Diameter of boom thread undercut	Gage
113.	Diameter at rear of spike thread	Gage
114.	Diameter of fuze cavity	Gage
115.	Diameter across wire slots	Gage
116.	Depth of annular groove	Gage
117.	Depth of spike thread cavity	Gage
118.	Length to inside basic diameter	Gage
119.	Length of thread boss	Gage
120.	Length of effective spike thread, min.	Gage
121.	Length total, max.	Gage
122.	Location of annular groove	Gage
123.	Thickness of wall at rear of annular groove	Gage
124.	Variation in wall thickness (indicated area)	Gage
125.	Width of wire slots	Gage
126.	Width of boom thread undercut	Gage
MINOR		
201.	Diameter of spike thread chamfer	Visual (See 6.1)
202.	Diameter of outside lead chamfer	Visual (See 6.1)
203.	Radii or chamfers missing or incorrect (except as otherwise classified)	Visual (See 6.1)
204.	Surface finish improper	Visual (See 6.1)
205.	Burr	Visual
206.	Foreign matter	Visual

4.3.2.5 Fin, Drawing D8597473, covering a detail of Drawing D8597472.

CRITICAL		
1.	Metal defective	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. True position of hub with pitch diameter of boom thread and front face	Gage
102. True position of the hub with the minor diameter of rear thread	Gage
103. True position of tracer cavity with the minor diameter of rear thread	Gage
104. Pitch diameter of boom thread, max.	Gage
105. Minor diameter of boom thread, max.	Gage
106. Length of effective boom thread, min.	Gage
MINOR	
201. Length, total	Gage
202. Radii or chamfers missing or incorrect	Visual (See 6.1)
203. Surface finish improper	Visual (See 6.1)
204. Burr	Visual
205. Foreign matter	Visual

4.3.2.6 Boom, Drawing D8597474, covering a detail of Drawing D8597472.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Pitch diameter of fin thread, min.	Gage
102. Major diameter of fin thread, min.	Gage
103. Diameter of forward basic length	Gage
104. Diameter of rear basic length	Gage
105. Length of fin thread pilot	Gage
106. Length from front face to fin shoulder	Gage
107. Length, total	Gage
MINOR	
201. Diameter of fin thread undercut	Gage
202. Width of fin thread undercut	Gage
203. Radii or chamfers missing or incorrect	Visual (See 6.1)
204. Surface finish improper	Visual (See 6.1)
205. Burr	Visual
206. Foreign matter	Visual

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4.3.2.7 Fin and Boom Assembly, Drawing D8597472.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Weight	Approved Scale
102. Runout of outside diameter of fin blades with pitch diameter of body threads and front face	Gage
103. Runout of outside diameter of boom with pitch diameter of body thread and front face	Gage
104. Straightness of fin blades	Gage
105. Pitch diameter of body thread, max.	Gage
106. Pitch diameter of body thread, min.	Gage
107. Minor diameter of body thread, max.	Gage
108. Pitch diameter of tracer thread, max.	Gage
109. Pitch diameter of tracer thread, min.	Gage
110. Minor diameter of tracer thread, max.	Gage
111. Diameter of counterbore forward of body thread	Gage
112. Diameter of counterbore rear of body thread	Gage
113. Diameter of tracer cavity	Gage
114. Diameter over fin blades	Gage
115. Diameter of fin hub	Gage
116. Diameter of tracer thread undercut (if applicable)	Gage
117. Depth of counterbore forward of body thread	Gage
118. Depth from rear face to bottom of tracer cavity	Gage
119. Depth of counterbore rear of body thread	Gage
120. Length of effective body thread, min.	Gage
121. Length of effective tracer thread, min.	Gage
122. Length, total, max.	Gage
123. Angle between fin blades	Gage
124. Thickness of fin blades	Gage
125. Thickness of wall between tracer cavity and rear thread	Gage
126. Width across top portion of fin blades	Gage
127. Width of tracer thread undercut (if applicable)	Gage
MINOR	
201. Depth from rear face to tracer cavity	Gage
202. Presence and dimensions of relief diameter	Visual (See 6.1)
203. Radii or chamfers missing or incorrect	Visual (See 6.1)

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4.3.2.7 (Cont'd)

	Defect	Method of Inspection
204.	Surface finish improper	Visual (See 6.1)
205.	Protective coating inadequate or damaged	Visual
206.	Burr	Visual
207.	Foreign matter	Visual

4.3.2.8 Wire and Insulator Assembly, Drawing C8597480, covering a detail of Drawing F10535894.

CRITICAL

None defined

MAJOR

101.	Diameter of wire holes	Gage
102.	Diameter over wire insulation, min.	Gage
103.	Diameter of hole in eyelet terminal	Gage
104.	Diameter of insulator head (large)	Gage
105.	Diameter of insulator head (small)	Gage
106.	Diameter of forward cavity	Gage
107.	Depth of forward cavity	Gage
108.	Length from front face to forward face of rear cavity	Gage
109.	Length of insulator head (large)	Gage
110.	Length, total	Gage
111.	Components damaged or defective	Visual/Manual
112.	Improper assembly	Visual/Manual

MINOR

201.	Excessive flash (insulator)	Visual
202.	Burr	Visual
203.	Foreign matter	Visual

4.3.2.9 Spacer, Fuze, Drawing C8597475, covering a detail of Drawing F8597471.

CRITICAL

None defined

MAJOR

101.	Height, total	Gage
102.	Length of tab	Gage
103.	Width of tab	Gage

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4.3.2.9 (Cont'd)

Defect	Method of Inspection
104. Thickness	Gage
105. Hole(s) missing	Visual
106. Metal defective	Visual

MINOR

201. Diameter (inside)	Gage
202. Diameter (outside)	Gage
203. Gap	Gage
204. Mismatch	Visual (See 6.1)
205. Width of slot above tab	Gage
206. Burr	Visual
207. Foreign matter	Visual

4.3.2.10 Conduit, Drawing B8597477, covering a detail of Drawing F10535894.

CRITICAL

None defined

MAJOR

101. Thickness of wall	Gage
102. Material defective	Visual

MINOR

201. Diameter (inside)	Gage

4.3.2.11 Projectile Sub-Assembly, Drawing F10535894, covering a detail of Drawing F8597471 (prior to paint).

CRITICAL

1. Metal defective	Visual

MAJOR

101. Weight	Approved Scale
102. Runout of pitch diameter of nose thread with bourrelet and shoulder below nose thread	Gage
103. Runout of spike stem with bourrelet	Gage

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4.3.2.11 (Cont'd)

Defect	Method of Inspection
104. Runout of large diameter of spike with bourrelet	Gage
105. Runout of crimping groove diameter with bourrelet	Gage
106. Runout of rear taper of body with bourrelet	Gage
107. Runout of pitch diameter of boom thread with bourrelet and rear shoulder	Gage
108. Runout of cone with bourrelet	Gage
109. Perpendicularity of large spike face with bourrelet	Gage
110. Diameter of spike flange	Gage
111. Depth of fuze spacer slot	Gage
112. Width of fuze spacer slot	Gage
113. Gap between body and spike	Gage
114. Length of fuze wire	Measure
115. Improper assembly	Visual/Manual
116. Threads damaged or not full	Visual
117. Burr	Visual
118. Foreign matter	Visual
MAJOR	
201. Length, total, max.	Gage
202. Terminal not properly crimped	Visual/Manual
203. Conduit hole in cone not properly aligned with slot in body	Visual

4.3.2.12 Metal Parts Assembly, Drawing F8597471 (after paint).

CRITICAL	
1. Metal defective	Visual
2. Air pressure stamp missing (indicating failure to pass air test)	Visual
MAJOR	
101. Pitch diameter of nose thread, min.	Gage
102. Pitch diameter of nose thread, max.	Gage
103. Major diameter of nose thread, min.	Gage
104. Pitch diameter of boom thread, min.	Gage
105. Pitch diameter of boom thread, max.	Gage

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4.3.2.12 (Cont'd)

Defect	Method of Inspection
MAJOR	
106. Major diameter of boom thread, min.	Gage
107. Diameter of bourrelet, max.	Gage
108. Paint in cavity inadequate or excessive	Visual
109. Foreign matter in cavity	Visual
MINOR	
201. Exterior paint not dry	Visual/Tactile
202. Paint in cavity not dry	Visual/Tactile
203. Paint on threads (excessive)	Visual
204. Paint on rear shoulder, beyond limits	Visual
205. Protective coating inadequate or damaged	Visual
206. Silicone compound missing from threads	Visual
207. Protector caps missing or damaged	Visual
208. Marking missing, incorrect or illegible	Visual

4.3.2.13 Ring, Locking, Drawing C10520021, covering a detail of Dwg. F10538894.

CRITICAL	
None defined	
MAJOR	
101. True position of inside diameter with major diameter of thread	Gage
102. True position of thread pilot diameter with major diameter of thread	Gage
103. Pitch diameter of thread, min.	Gage
104. Major diameter of thread, min.	Gage
105. Diameter of thread pilot	Gage
106. Diameter (inside)	Gage
107. Length of thread	Gage
108. Metal defective	Gage
MINOR	
201. Length, total	Gage
202. Presence and dimensions of wrench slots or holes (as applicable)	Gage

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4.3.2.13 (Cont'd)

	Defect	Method of Inspection
203.	Radii or chamfers missing or incorrect	Visual (See 6.1)
204.	Surface finish improper	Visual (See 6.1)
205.	Burr	Visual
206.	Foreign matter	Visual

4.3.2.14 Plug and Disc Assembly, Drawing B8597489, covering a detail of Drawing F8597471.

CRITICAL

None defined

MAJOR

101.	True position of inside diameter with major diameter of thread	Gage
102.	Pitch diameter of thread, max.	Gage
103.	Pitch diameter of thread, min.	Gage
104.	Major diameter of thread, min.	Gage
105.	Thickness of disc	Gage
106.	Thickness through body	Gage
107.	Improper assembly	Visual/Manual
108.	Metal defective	Visual
109.	Foreign matter	Visual
110.	Threads damaged or not full	Visual

MINOR

201.	Diameter of thread chamfer	Gage
202.	Distance between wrench holes	Gage
203.	Surface finish improper	Visual/Measure
204.	Burr	Visual

4.3.2.15 Pallet, Drawing F8597604 for Metal Parts Assemblies.

CRITICAL

None defined

MAJOR

None defined

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4.3.2.15 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Strapping missing, loose or broken	Visual/Manual
202. Pallet broken	Visual
203. Improperly packed	Visual

4.3.2.16 Cartons, prior to sealing.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Support missing or damaged	Visual
202. Separator missing or damaged	Visual
203. Improperly packed	Visual
204. Incorrect number in carton	Visual
After sealing.	
205. Contents loose	Manual
206. Carton damaged, cut or gouged through all layers	Visual
207. Carton not completely sealed	Visual
208. Marking missing, incorrect or illegible	Visual

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level C - Minimum military pack.-The metal parts for Projectile 76mM, HEAT-T, M495, shall be packed as follows:

5.1.1.1 Metal parts assembly - Part no. 8597471.-Shall be packed in accordance with Drawing F7548604.

5.1.1.2 Fin and Boom assembly - Part no. 8597472.-Shall be packed in accordance with MIL-STD-1169. Pack 25 fin and boom assemblies arranged in a 5 x 5 pattern, ussing interlocking separators.

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5.1.1.3 Spacer, Fuze - Part no. 8597475.-Shall be packed in accordance with MIL-STD-1169, in quantities of 1000.

5.1.1.4 Plug and disc assembly - Part no. 8597489.-Shall be packed in accordance with MIL-STD-1169. Pack 2500 Plug an disc assemblies arranged in a 10 x 10 pattern, 25 to a cell, using interlocking separators.

5.1.1.5 Percentage of extra parts packing.-Packing for the required percentage of extra parts shall be provided for by furnishing the proper number of prime containers.

5.1.1.6 Shipping.-When component parts from more than one lot are shipped as a carload, each lot must be separate and the division between lots clearly indicated.

5.1.1.7 Marking.-The pallet shall be marked in accordance with Drawing F7548604. Cartons shall be marked in accordance with MIL-STD-1169.

5.1.1.8 Data cards.-Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirement is in doubt as a result of a visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Visual standards.-Visual standards will be established at contractors plant in compliance with 3.7.3 or whenever directed by the Procuring Contracting Agency.

6.3 Alternative testing procedures.-Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisons specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible technical agency that their effectiveness is equal to, or better than, the quality assurance provisions specified herein.

6.4 Hydrogen embrittlement.-After initial production sample acceptance and after the contractor has presented evidence to the procuring agency concerning the adequacy of his furnace atmospheres used in annealing, the contractor may request that testing for hydrogen embrittlement be discontinued. The decision regarding elimination of testing for hydrogen embrittlement will be made at the discretion of the Procuring Agency based on the results of the initial production sample inspection and on the evidence presented by the contractor concerning his annealing equipment.

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6.5 When a product is being inspected in accordance with MIL-STD-1235 and a defective characteristic is found, product produced prior to the discovered defect will be screened until the frequency (f) is found to be free of that defect.

6.6 Document availability.-The Appendix of the specification is a classified document and will not be distributed through normal channels. Copies may be obtained on a need to know basis through the Commanding Officer, Frankford Arsenal, ATTN: SARFA-TSA-T, Philadelphia, Pennsylvania 19137.

6.7 Ordering data.-Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1315-A253

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